

ActInf Textbook Group ~ Cohort 4 ~ Meeting 20

(Chapter 9 part 2)

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all right hello everyone welcome back it's our second discussion on chapter n so we can look at the chapter and look at previous questions but before we do so does anyone just want to give any general comment or thought or question on nine like any section of nine that was interesting or any topic sorry I'm running a bit behind in the reading so uh haven't looked at chapter n or or even another topic or another question if you want to go to it that's sorry yeah go for itna oh no I just I'm this is my first time attending and I did not know that we were going to be working on chapter nine so now that I'll be I'll prepare but I was focusing uh I focus my work mainly on making sure that I understand some key Concepts in the book and so so I'm not really quite ready today oh it's all good thank you for joining I mean it's always listed here but also it's kind of like we even talk about things outside the textbook so we can talk about whatever people want even just the theme of chapter nine with um modelbased data analysis I mean this is the chapter where the models are connected to empirical data that's always a useful topic or or just anything people want to talk about or we can look at some prior questions I the only thing just a really minor comment about the book uh I just noticed that Parts one and two are not don't have labels on the index page that's that's uh that's one thing that I picked out bun um but like I said I'm I'm new to this topic I will get I will do a lot of reading this week but I really wanted to make sure that I understood where the free energy principle concept came from so I'm really way behind so it's all good it's all good um when you say part one and part two you mean chapters 1 through five and chapter 6 through 10 uh well when I have the PDF and on the PDF on the first on the index page I not is that that Parts one and two there no there's no label it just has the list of the sections like this let me see part one and part two part one and part two y on the on right on the in the content page yeah so the label the the label of the labels of the parts are not in right on included and I didn't know if that was deliberate or yeah I think the part one and part two is just kind of like an informal okay like the chapters are the main components yeah um and if you want to know like for the origin

the fvp chapter 10 might give you some information oh good okay it's yeah okay any other chapter 9 related topics um or Andrew had some things to discuss or anyone else could bring something up yeah or go for it yes please oh okay thanks Daniel thanks Angel um yes I have a couple questions regarding chapter uh um so um I wonder how should I uh share with please add them to the questions everyone is just overwhelmingly encouraged to add the written questions to the table write questions uh in the table yeah where make a blank row and just put it into the question part okay okay um yeah let me um okay so I have to go to this uh this website right to share the questions yeah you can follow like the the you can click like the token for active inference to follow to where I am in chapter nine yes I find chapter nine yeah so add add the questions there okay okay then let me check if I'm on the same page as you do yeah uh chapter nine yeah yeah yeah I'm I'm there so so okay okay so I just juggle between U so first um I have a question regarding uh regarding equation 9.2 um so so what is um this $\Theta \alpha$ in equation 9.2 and similar writing did you enter the question or where did you write okay okay what is I'm gonna I just copied it into there but yeah oh at the wrong place okay no no it's okay it's okay just this is how we can get all the questions together what is $\Theta \alpha$ so because it hasn't show up in our previous you know chapters before right this $\Theta \alpha$ sub α yes Θ sub α yeah yeah okay so um just to take a step back we're using variational Lelo approximation so what this technique is doing is it's taking some distribution it's going to find the central tendency of it and then it's going to fit an inverted Parabola over it and so that could be a narrow parabola Parabola or it could be a wide Parabola so that's the variational Lao so it is a lot like kind of fitting a gaussian approximation but of course a gaussian has support everywhere and it continues whereas a parabola like just goes down um so that's what the lla approximation does Θ is being used as commonly to just refer to the total set of parameters okay okay now um down here this is going to be in the context of policy selection so there's going to be beliefs about action sequences we can take these beliefs and calculate the average probability of pursuing an action sequence so this is in the spirit of pursuing action selection as a pure statistical problem and finding the most likely path path of least action not appealing to any kind of like intermediate reward function or anything like that this requires us to distribute the probability for each policy over the actions implied by that policy so the array U is going to connect the actions or the affordances to the policy actions or affordances are things that can happen in a single time step policies are sequences of multiple time steps in length so it's basically like saying each policy was evaluated so time Step n and then um each policy is described in terms of what actions it

contains finally a soft Max temperature parameter Θ_{α} is applied to account for Randomness or shaky handedness in Behavior not accounted for by the model so this is kind of just brought up here but it's explored more in other papers um when the soft when the softmax is one your behavior is sampled directly from your policy posterior that's matching behavior when the soft Max is very high then you select the best behavior even if it's only slightly better and then if your soft Max is very low it basically erases differences in the posterior likelihood of action okay so so Daniel this this uh it's new parameter is not needed in the um in the active inference model right it's only needed when we um when we do this parameter estimation right when we compute the likelihood function right I mean the core active inference model is very sparse it doesn't contain learning or attention or shaky hand so the textbook is showing us all these different motifs none of them which you have to do but if your phenomena of interest is captured by this type of behavior this is a useful Motif but it's not like any of them are being required for us to do in all situations it's just saying this is one kind of um structural model that can capture decision-making Behavior okay okay but you're you're right like previous examples didn't have this right right I I also would like to clarify the policy you just mentioned there so so policy um you said the policy specifies actions uh at each time point if we think about the discrete time model right um so it simply says that know what action I would take at time one what action I would take a Time um yep here I just went to the ontology just looked up policy it's a sequence of actions okay I wonder because in in my area where I think about policy as a as a mapping from state to action uh not simply from time to action so you're you're thinking about about you know policies in time one what I do in time two what I do uh in general uh a policy defined as in if in certain State what I should I do time just is one dimension of the states um is this a a special feature of active inference or is it's something like consistent with the other like Dynamic uh process it's more similar than not to how policy selection is discussed in reinforcement learning deep Q learning because fundamentally we're talking about action selection so it's overwhelmingly more similar than not but yes there are some slight differences because here we're only talking about policies as sequences of action through time and then using other features of the model like other state um inferences of the model to update our posterior on policy but we're we're not we're not making policy explicitly a function of other states other than time yeah it's generally the same control theory setting with just a slightly different setup but but you you should document what is meant by policy in that setting pull out the quote or pull out how it's meant and then let's just just toose it side by side and then it'll be clear right in control theory in general the policies is

the action is function not only the time but also the system States right that's right the sequences of actions are a function of parameters the model and a sequence of stimuli that's here the likelihood of the parameters is describing the sequence of actions Through Time conditioned upon parameters model and observations we can go more into it but it's it's it's the same setting of inferring actions from other features of the model okay so what you what you observes are this actions okay yeah um here here's Us in the behavioral laboratory observing U Tilda the object empirical behavior of the system of Interest objective here it's a objective meaning empirical like what we objectively measured and recorded empirically in our laboratory so what we recorded as data is the output of the subject of our behavioral experiment so this isn't taking on the the the the philosophy of you know objective and subjective and all this this is just a behavioral setup here's the rat that's our behavioral subject it has a cognitive Model A generative model and we're observing Its Behavior and that is being used as our empirical scientific data for cognitive modeling so o is also input by the uh experimental designer by the um so it's it's given kind of a parameter right the stimuli yeah the experimental stimuli is a action by the experimenter and a sensory State for the subject yeah I think that the the key here that utia is the output of the experiment the behavioral experiment uh and the input of the the fitting process that gives you the Theta once you have done the measurement UA you can guess which parameter is the is the best one to describe this vavle right right yeah here is the G action to find the best parameter theeta right this what the uh or it's the purpose Theta is the parameters um so so the other question I have actually in this figure 9.1 is what is m so it seems M just the model all the parameters of the model then why why do we use like different symbols one use M another use Theta M for the generative model of the of the subject and Theta for the experimental parameters okay for example in the T example chapter seven what is M and what is what is Theta what are the differences M would be the parameters that describe the cognitive model of the Rat Theta for the parameters describing the setup of the teamas like the idea that 50% of the time the food is over here or that when the food is here it has a 99% probability of reward okay so M basically represents other parameters not uh this is just a schema you know this is just saying yes this it's you don't have to use exactly this one and only basian graph so it's just saying whatever is the subjective model you can struct is you're going to be conditioning upon the experimental parameters and the cognitive model parameters okay exper um so Theta is experiment experimental parameters so those are the parameters that we are interested in right and they might be some of the ones you're interested in but different Laboratories are interested in different questions

some might have a very trivial subjective model and be interested in doing inference up here other times this might be fixed up here and there might be something interesting here may I interject a question yeah yeah okay so so Daniel something I I been thinking about about that's this is more of a philosophical question but I really like this representation uh I just wondered if it matters in the free energy principle approach to be worried about nested cognitive systems um because in the work that I'm doing nestedness is uh for the system that I'm describing there are subsystems so we have a system within another system within another system so from a very abstract point of view the nestedness is that not relevant because at every boundary you're just dealing with a different set of sensory input from something else so it doesn't matter if you're missing anything you're just a system and you can model the behaviors of that system separately from the fact that it's receiving input the inputs are coming from multiple levels of higher order that affect it does that make sense yeah I mean nested models are are naturally amenable to this type of modeling okay so here's a nest an example of a nested model so here like we we see this kind of figure 4.3 generative model shape um now that that like e downstairs like a downward facing e that's like the sense making component and then the upstairs with a policy selection so now that is kind of expanded so that each Leaf of the E has another nested e and so there's nothing in active inference that says that systems must be nested however it is very amenable to modeling Collective lateral interactions within a level and then having nested models across levels and then just briefly two kinds of nested models that we see sometimes the nested model refers to an actual spatial enclosure so you know a continent and then a region within a continent and then a village within the region um with like slower and bigger things being at higher levels but also we see nested models for metacognition where you have higher levels of metacognition as nested observers and and top-Down actors of lower level of cognition so it's not spatially enclosing but it could be thought of as like a cognitive enclosure I understand that yes that that was very helpful thank you uh but so let me see if I can ask this a little this is very abstract but um is it the case that even so even when we're looking at those nested models um The Nest the sub models the the ones that are nested within a bigger uh model there are there any modeling issues that you need to add to to your subsystem in order to take into account that it's receiving sens uh inputs from multiple higher order systems or does it or you just U from a modeling perspective it doesn't matter you just look at yes they look identical in principle it's well behaved in practice the challenge is that nested models get very very large very quickly so then it it's like well you know I mean you know the pancreas is nest in the liver and the liver and the chest and the

person and the family and it's like all of a sudden you're just with a model with with millions of degrees of freedom with limited data sets so a and yet making a a appropriately articulated nested model actually can help the model embody the structure of the world which uses the data more effectively so um a smartly constructed nested model will help you make the best of the data you have but a sort of nested model construction as a fishing Expedition ends up losing interpretability even if it ends up recapitulating real features um yes so no that that's that thank you that was super helpful because actually so one of the boundar Aries to just to give an example one of the boundaries that I'm working with is the boundary between or or thinking about um the the human body and um the human body under conditions of our early evolutionary adaptedness environment and what's really interesting about humans is that we have this um um brain got aess cyc neuroimmunological sophisticated system that has a very uh well documented effect on the brain uh when uh the immune system is reacting to a pathogen so it sends signals to the brain and causes what's called um sness Behavior right so that's an example of communication that we understand that's going on between subsystems and the Brain um but we actually can talk about the communication the way that the communication is happening so you don't have to be doing a fishing Expedition um you you know eventually I would like to be able to model those relationships in some way yeah that's a great transition actually to what um Andrew um brought up last week so like since we're in chapter nine um you know we we're talking about the empirical data but but the book only has limited amount of empirical data but it cites um heavily these other studies but then Andrew suggested that we go into this one so I mean Andrew what what did you want to discuss about this paper I pulled out some of the the um figures that we can talk about but what do you want to talk about it sure um see I had a limited amount of time available between last week and this week but the the initial thought was um just kind of because this chapter lays out kind of a series of steps that one could follow as a sort of recipe um the idea was maybe we could um identify in in layman's term well as far as we can in layman's terms like um something like what are what are those steps as played out like for example in this paper here um yeah so we kind of have like we can match like to each step of the recipe provided like maybe what's going on here right awesome yeah so chapter six we constructed the generative model um here and these are all like steps that have to happen but this isn't exactly needing to happen this order but chapter six we basically construct the um generative model that's step two and it's Associated likely Ood function which is just the function of behavior being emitted conditioned upon parameters observations in the generative model like just that we talked

about with 9.2 and then we collect data we crunch it all together invert the basian model and then do statistics on the outcome so let's see what it looks like in um the case of Ryan Smith at all's work on gut inference so they develop a asian computational model for gastrointestinal interoception so this is not exactly like the microbiome neurotransmitter slower time scale interaction but this is the faster like vagal nerve mediated interception with the gastrointestinal um so it was completed by healthy individuals they were undergoing EEG so there was getting neuro neuroimaging data and peripheral physiological recording like C like some measures of the body and then this I also found this kind of interesting these self-report questions how hungry do you feel how you know how difficult was this task this is just showing like you can assess a variety of physiological measures self-report measures these are all data okay okay another important thing that that to keep in mind is like it's not about constructing the one generative model it's about having a portfolio of related generative models that do and don't have different components and then you kind of test across them um like when they talked about what are the two reasons to do cognitive modeling one of them was if you have a fixed generative model then you can look for group differences another one is from the data set you can identify which models are appropriate so here these 12 models differ kind of combinatorically in their different features so standard statistical modeling technique test for you know just Mo factor a just Factor B A and B together A and B and C just C and D together so so that's what they they're going to um do and in table one they Define the same ontology terms that we see in the textbook um here's the test they swallow this pill that vibrates in their stomach and then they signal whether they did or didn't perceive it so the true State hidden state is whether it vibrated or not that's known because that's controlled by the experimenter um and then the observations are the perception that they experience obviously if it was a fully observable situation you'd have like the a matrix would be an identity Matrix you would just simply be able to perceive it but there's some intensity of signal where what they're going to test is that well it turns out that it's not 100% perceived but also it's not 0% perceived and maybe there are interesting patterns of people's perception here is literally where we see the basian graph now this is a model that doesn't include action because the person is not in control of vibrating this little thing so we only are seeing the downstairs part of the model so in that way it's um much like the early example in 7 so this was the music listening example in chapter 7 just the downstairs the prior hidden State Through Time observations at each time step but no upstairs action selection component prior hidden State on the actual vibration observations whether it's perceived a is the mapping between vibration State and observation

B is how it's transformed so that model could be specified ahead of time you could pre-register the study so you kind of know that you want that and then they describe this a lot more in the caption um they also describe yeah so then they do some descriptive statistics on this model the interceptive Precision so that's the Precision of the a matrix if the a matrix is just the identity matrix it's the most precise that means that you have perfect observation if the a matrix was low Precision it was just a wash you wouldn't be able to perceive anything and then it could be anything between um prior beliefs favoring the presence of a vibration PV that's the D variable and then the learning rate is describing the the rate of change on B so those are not simply descriptive features of the empirical data if you just wanted to say who's has a higher accuracy or who's more right or something you could just summarize the data you wouldn't even need to do cognitive modeling then they look at all of these different empirical patterns so here on the columns these are um Prima fascia components of the behavioral study these are just descriptive outcomes from what empirically did happen like the true positive negative rate and the reaction time patterns on the rows are these cognitive parameters that again they identified with their model so features of their model map not the territory and yet there are still these interesting patterns um then they they can do different conditions and show that their the cognitive parameters that they're interested in that those cognitive parameters are variable in different settings here they look at the the cognitively inferred parameters on the rows and then a a uh self-report and some kind of like um personal data aent BMI um just like other scientific studies like the kind of result result is like a point estimate and a variance so you can say some of these results are significant at different thresholds now they're going to connect the cognitive model parameter to EEG Dynamics here they look at clust string of erps event related potentials so these are kind of like dynamical patterns in the EEG over different time scales and then they can cluster different regions by the similarity of their response and look at how those kind of traveling waves are related to um uh those traveling waves the erps are related to those different cognitive modeling and they do that for other brain regions too me then D can I ask a question yeah the the um corination Matrix showed uh sorry the before with the learning rate is Nar rate estimated from the uh data yes everything is estimated from the data in this paper okay okay so so how how do they construct the magn function to do the estimation we use the um EG data to to estimate I mean you look look at the paper for more details but um the EEG data are being taken as kind of like this is you know what you see is what you get here right there's no EEG here there's just a prior on vibration and then an an observation and then the mapping between the observation and the vibration or not so that's

kind of like the core phenomena that they're studying and then they have the age and the BMI and the EEG and the self-report so then those other data are kind of overlaid on the timeline of this kind of core model and then you can look for correlations [Music] between um this graph and the EEG so I don't think you need to take in the likelihood of the EEG data specifically well but in order to estimate at the learning rate they do need a MAG function right so it has just parameter of this active inference model yeah I mean we can look for more details on how the learning rate is calculated but they describe it in that caption I mean it's it's described but yeah look into it and see if that's how you would describe learning rate and these are features of the map not the territory so again it's common for people to look at this and then reference some feature of the world but it's not about referencing some feature of the world to validate or invalidate this model it's about generating a portfolio of models that have different features and then propagating that portfolio of models forwards and then saying actually learning rate does matter because the one that had it versus the one that didn't had this better fit and someone says oh but you didn't think about this and you go okay cool now let's add model 13 with that feature and that's the real that's the Delight of the empirical conversation just people bring things up and it's like okay either we have explored it and we can talk about the impact of that variable or we haven't considered it yet so then thank you for the suggestion let's add it in and then come back to it and that's the iterated process but yeah here by looking at the table how how do we know which model is a better description um so you have 12 models to yeah yeah yeah great question they they describe this the most common method used is the Bayesian Information Criterion BIC or the related um AIC um the BIC criteria basically um rewards fitting a model uh fitting the data well and it penalizes having extra parameters and so because you're adding every model with another parameter you will always get a better fit like a principal component analysis the next principal component all always is going to soak up more variance so make the best fitting model is actually a kind um ironically useless singular goal but if you say I want the model to be at the optimal tradeoff where like the parameters are like punching above their weight so that's usually how people select it but that's not the only model selection imperative you might say I'm okay with having a huge model I just want the most accurate one so then you might take one that has more parameters or you might say I don't need the most accurate one I need it to just be better than 50% and I need it to you know fit on an index card yes there isn't an answer in the general case for which model you should select and you might want to actually just hold the whole portfolio and just keep that whole distribution that's like Ensemble modeling yeah so so I'm just wondering why

it doesn't show the selection criteria in the table when they selection criteria yeah yeah so the BC they didn't show us right um well not in that table like I I mean I the the models are compared in some formal way but in that image they're just they're just listing which ones they are but yeah it could have included like the likelihood of the Bic for the models or the number of um it's just not shown in that representation maybe in the supplement or maybe somewhere else but but yeah that that's usually what is shown you see like it will be like model one and it will be like Delta Bic Z and then it will show how how much behind in the Bic the other ones are um and then if we were studying an active setting like let's just say that now we're in a little modified version where the person can make can be in the you know red or the orange room and they can move between those two rooms and then those two rooms have like a different probability of the stomach thing vibrating so now there's some variable well that's the power of the base graph it's just another parameter to estimate so the same ex and then someone says well what about having a policy on the air conditioning as well then you add the air conditioning policy oh and now we want the observations not just to be the interception now we want to have like you know exter reception too you just keep composing and building the models but in chapter nine that's where we make the contact with the empirical and that kind of like reminder that there are real things to study and that you can design the generative model ahead of time and then do the experiment and then just as soon as you collect the data you just dump it into the generative model and you basically have your paper ready that's one great option another option is the data have already been collected and you know what structure they are and then you can construct a generative model that will load in that structure so whether the data are collected before or after obviously it's it's different scientifically but in both cases this chapter gets us to the point where we can take in data bring it into this kind of format but then in the end we make the kind of bar bar plots and time series and correlation matrices that we would see like from literally any other scientific paper yeah what what did anyone think about this well I'm just kind of stepping back and I wanted to uh ask you uh so is a lot of the literature that on recent literature on the energy principle about you know when we going to the empirical world is figuring out or trying to infer what generative models different systems have in order for us to be able to begin to test hypothesis from the free energy prot that's a great question yeah definitely like one line of research or multiple lines of research are related to the fundamental physics of cognition some lines of research are more computational capacity building and so they're interested in like motifs and comp computational implementations another set of research that's more empirically oriented is looking at

empirical data and developing models that's one direction of empiricism and then the other direction of empiricism is with the cognitive model in hand looking at differences amongst individuals or groups and like Ryan Smith ated all's work has both of those directions of empiricism sometimes um like in this paper they're taking a simple setting and then they're exploring a range of models in other settings they're basically taking a cognitive model identified from previous work or however and then asking okay do individuals with these different diagnoses differ for example and those those two directions of empiricism can coexist in one paper but but then it's kind of like this joint identification of what should the model be but then how do the individuals differ in the model and here's those two reasons that they say computational phenotyping with the model in hand describing individuals or with the data in hand ask which model you want to utilize Daniel can can I ask you a question here yeah about T's example um in chapter seven the textbook provides a complete example about the uh active influence model and in this chapter um they also talk about that t example but I wonder if there is any empirical data about the the mouse you know movement and that can we can like you know use the same example to illustrate the the basic steps and idea nine yeah I mean it's a great question something I've wondered about too don't we have like hundreds of thousands of data points on the teamas I don't know what data set that is but but these are experiments that have been done here those are simulated data not the experimented data right yeah here they just were using the teamas in the synthetic Direction but yeah it's a great project to find an empirical data set um and then you would need to make the generative model for it because it wouldn't be exactly this but this structure might work well but yeah I mean this is the work that I hope that the textbook gets us all to is going to behavioral settings and studies that have been done and building generative model but this is literally the work that we have to do well you're you're not aware of anyone have done it right because I I asked a few people who who uh extend this uh teamas uh example and but none of them have done empirical um study and estimate The Motto parameters I've never seen it with like the mammal Tas again even though they're they're are actually probably thousands of data sets on the Tas so I think that's a a very viable option perhaps people use the teamas because it's easy to make a point about epistemic value but usually the teamas isn't the case that people are interested in studying but if it were a lab that were studying decision making or addiction or learning in rodents then they would be actually looking to do a cognitive model with the teamas so that's where the collaboration between people with active inference expertise and domain experts and then you have an iterated modeling process with them are all the features

included in this model and the back and forth there and then find out what the generative model what kind of empirical insights you can have and in that case if it's already a data set that's collected in a published paper that's the perfect setting because then you can say here's what active inference goes beyond descriptive statistics and what it actually what value it does add to the setting yeah yeah there's a lot of Behavioral data sets to do this on yeah if you were actually looking for data sets I mean there's a great um the busak labs there's um Yori rousaki neuroscientist at NYU he has an entire lab dedicated to um things like for example studying like mice and and rats are kind of his expertise and um you know running many experiments where they're trying to use um electrophysiological data to track uh what is going on in mice what's going on in their their neurons whenever they're navigating a maze and it's where they do research on things like um Place cells and all these fascinating things where if you draw out a topological map out of the EEG data you can actually see like the the the mouse is tracking where it is location so I see Daniel's point to bring it back that there are many experiments done that highly relate to the tas's example but I guess it's around there that you could have collaboration between someone who's an expert in active inference versus those who carry out these kinds of experiments to test the data that's already been collected uh but within more of active inference kind of framing because because the the the lab technician here at busak or otherwise are attempting to understand something more like the neurobiology but they're not necessarily getting too indepth with the cognitive modeling aspect or the kinds of questions that someone um doing research in active inference might be asking that's my thought on that right yeah thanks angel so um this is a uh this is a question that is bugging me and I just want to bring it up now um before I get really deep into all the material which is um when working with human subjects uh you know one of the things that that that I see as a is it's always been a huge problem in out my research is that as soon as you enter into the interview setting that in itself you know it has a huge observer effect so so from the perspective of active inference you're dealing with the fact that this individual from whom you're getting some information about their generative model in their heads they have a generative model in their heads about that interaction and making and already trying to make assumptions or thinking about what is this interviewer going to ask me and what are their expectations because that's what we humans do and so if you could tell me early on where in this world of active influence is that addressed because it's a huge issue in in my systems in the systems that I work with yeah it's a big question there's a lot lot of pieces to say on it okay um suffice to say that studying like a pendulum is an easier active system than study an active system that has theory of mindly

okay thank you thank you that was like really well but so yeah it's not to be found in this image but this image is kind of like yeah pointing towards it okay yeah thank you to the textbook group um we return for chapter 10 next time okay thank you very much take care bye